

exercice n°2 :

exo 01:

Let:

$$f(x) = \tan x$$

$$f'(x) = \frac{1}{\cos^2(x)}$$

$$f^{(2)}(x) = 2 \left(\frac{\frac{1}{\cos^2(x)} \cos^2(x) + 2 \sin(x) \cos x \frac{\sin x}{\cos^3(x)}}{\cos^4(x)} \right)$$

$$= \frac{2}{\cos^4(x)} + \frac{2}{\cos^2(x)} \tan^2(x).$$

$$f'(0) = 1, \quad f^{(2)}(0) = 0, \quad f^{(3)}(0) = 2$$

$$f(x) = \sum_{k=0}^3 \frac{f^{(k)}(0)}{k!} x^k + R_n$$

$$\tan x = x + \frac{1}{3} x^3 + R_3(x).$$

Using Lagrange rest $R_3(x) = \frac{f^{(4)}(c)}{4!} x^4$

for some $c \in]0, \frac{\pi}{2}[$. we have:

$$f^{(4)}(c) = 8 \left(\frac{\cos^2(c) \sin(c) + \sin(c) (1 + \sin^2(c))}{\cos^5(c)} \right)$$

Hence:

$$\tan(x) = x + \frac{1}{3} x^3 + \frac{f^{(4)}(c)}{4!} x^4 > x + \frac{1}{3} x^3$$

because: $\frac{f^{(4)}(c)}{4!} x^4 > 0$, for some,

$$c \in]0, \frac{\pi}{2}[.$$

exo 02:

Using T-L formula to order 3

at 0 to proving the second inequality

$$f'(x) = \frac{1}{1+x}, \quad f''(x) = \frac{-1}{(1+x)^2}$$

$$f^{(3)}(x) = \frac{2}{(1+x)^3}, \quad f^{(4)}(x) = \frac{-6}{(1+x)^4}$$

so:

$$\ln(1+x) = \ln(1) + x + \frac{-1}{2!} x^2$$

$$+ \frac{2}{3!} x^3 + \frac{-6}{4!(1+c)^4} x^4, \quad c \in]0, x[.$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4(1+c)^4}$$

$$\leq x - \frac{x^2}{2} + \frac{x^3}{3}$$

2) using Taylor-Lagrange to order 4 at 0, for proving that

we have $f^{(5)}(x) = \frac{24}{(1+x)^5}$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{24}{5! (1+c)^5} x^5$$

$$\geq x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4}$$

so:

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} < \ln(x+1) \leq x - \frac{x^2}{2} + \frac{x^3}{3}$$

approximation of $\ln(2)$.

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} \leq \ln(2) \leq 1 - \frac{1}{2} + \frac{1}{3}.$$

$$0,58... = \frac{7}{12} \leq \ln(2) \leq \frac{5}{6} = 0,833..$$

2)

$$f(x) = f^{(1)}(x) = f^{(2)}(x) = f^{(3)}(x) = \exp(x)$$

$$\text{for } x=0 \Rightarrow f(0) = 1.$$

$$\text{so : } \exp(x) = 1 + x + 1 \cdot \frac{x^2}{2!} + \frac{x^3}{3!}$$

$$+ \exp(c) \frac{x^4}{4!} \quad (\text{for } c \in [0, x])$$

$$= 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \exp(c) \frac{x^4}{4!}$$

$$\geq 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}.$$

$$\text{Since } \exp(c) \frac{x^4}{4!} > 0.$$

$$f(x) = (1+x)^\alpha, \quad \alpha \in \mathbb{R}.$$

* Taylor formula to order n at $x_0=0$:

we have: $f \in C^n(\mathbb{R})$, so:

$$f(x) = 1 + \alpha(1+x)^{\alpha-1} \cdot x + \dots + \frac{\alpha(\alpha-1)\dots(\alpha-(n-1))}{n!} (1+x)^{\alpha-n} x^n + O(x^{n+1}).$$

$$f_1(x) = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2} x^2 + \dots + \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!} x^n + O(x^{n+1}).$$

$$1) f_1(x) = \frac{1}{1-x} = (1+(-x))^{-1}.$$

using Taylor formula of $(1+x)^\alpha$ we get:

$$f_1(x) = 1 - (-x) + \frac{(-1)(-2)}{2!} (-x)^2 + \dots + \frac{-1(-1-1)\dots(-1-n+1)}{n!} (-x)^n + O(x^{n+1}).$$

so:

$$f_1(x) = 1 + x + x^2 + \dots + x^n + O(x^{n+1}).$$

$$f_2(x) = \ln(1-x).$$

$$f_2'(x) = \frac{-1}{1-x} = -(1+(-x))^{-1} = -f_1(x).$$

so, by integration:

$$f_2(x) = -\left(x + \frac{1}{2}x^2 + \dots + \frac{1}{n}x^n + O(x^{n+1})\right).$$

$$3) f_3(x) = \frac{1}{(1-x)^2}.$$

we remark that $f_3(x)$ is the derivative of f_1 .

$$f_3(x) = 1 + 2x + 3x^2 + 4x^3 + \dots + x^n(n+1) + O(x^{n+1}).$$

$$4) f_4(x) = (1+x)^{-1}.$$

$$f_4(x) = 1 + (-1)x + \frac{(-1)(-2)}{2} x^2 + \dots + \frac{(-1)(-2)\dots(-n)}{n!} x^n + O(x^{n+1}).$$

$$5) f_5(x) = \ln(1+x).$$

Notice that: $f_5(x) = f_2(-x)$.

$$f_5(x) = -(-x) - \frac{(-x)^2}{2} - \frac{(-x)^3}{3} - \dots - \frac{(-x)^n}{n} + O(x^{n+1}).$$

$$6) f_6(x) = \operatorname{arctanh}(x)$$

$$f_6'(x) = \frac{1}{1-x^2}.$$

and let $y = -x^2$.

so:

$$f'(y) = 1 - y + y^2 - y^3 + \dots + (-1)^{n-1} y^{n-1} + o(y^n)$$

by integration:

$$f(x) = x + \frac{1}{3} x^3 + \frac{1}{5} x^5 + \dots + (-1)^{n-1} (-1)^n \frac{x^n}{n} + o(x^n).$$

$$f(x) = x + \frac{1}{3} x^3 + \frac{1}{5} x^5 + \dots + \frac{x^n}{n} + o(x^n)$$

$$\begin{aligned}
 &= 1 + \frac{1}{2} x^2 + \frac{\frac{1}{2}(\frac{3}{2})x^4}{2!} + \frac{\frac{1}{2}(\frac{3}{2})(\frac{5}{2})}{3!} x^6 + \dots + \\
 &\dots + \frac{\frac{1}{2}(\frac{3}{2})(\frac{5}{2})(\frac{7}{2})\dots(\frac{2n-1}{2})}{n!} x^{2n} + o(x^{2n})
 \end{aligned}$$

$$= \sum_{k=0}^n \frac{\prod_{i=0}^{k-1} (2i+1)}{2^k k!} x^{2k} + o(x^{2n})$$

$$f_{10}(x) = x^n \cos(x) = \int \frac{1}{\sqrt{1-x^2}}$$

$$= \sum_{k=0}^n \frac{\prod_{i=0}^{k-1} (2i+1)}{2^k (k!)} \frac{x^{2k} 2^{-2}}{(2k 2^1)} + o(x^{2k+2})$$

$$\begin{aligned}
 p_{02} &= \int f_{11} \\
 &= x + \frac{1}{8} m^3 - \frac{1}{40} m^5 \\
 &\quad + \frac{1}{2} m^4 \sim 13x \quad (2m^3)
 \end{aligned}$$

Exo 4:

$$f(m) = \frac{1}{m} \ln(2-m) \text{ at } 1, \text{ order 2}$$

we have 1GDg:

Basic technique

$$f(1) = 0$$

$$f'(m) = -\frac{1}{m^2} \ln(2-m) - \frac{1}{2-m} \left(\frac{1}{m} \right)$$

$$f'(1) = -1$$

$$f''(m) = \frac{1}{m^3} \ln(2-m) + \frac{1}{m^2} \left(\frac{1}{2-m} \right)$$

$$= \frac{1}{(2-m)^2} \left(\frac{1}{m} \right) + \frac{1}{m^2} \left(\frac{1}{2-m} \right)$$

$$f''(1) = 1$$

$$f(m) = -(m-1) + \frac{(m-1)^2}{2} + (m-1)^2 \mathcal{E}(m)$$

second technique

$$f(m) = \frac{1}{m} \ln(2-m) = g(m) \cdot h(m)$$

$$f(1) = 1$$

$$g'(m) = -\frac{1}{m^2}, \quad g'(1) = -1$$

$$g''(m) = \frac{1}{m^3}, \quad g''(1) = 1$$

$$g(m) = 1 - (m-1) + \frac{(m-1)^2}{2} + (m-1)^2 \mathcal{E}(m)$$

$$h(m) = \ln(2-m), \quad h(1) = 0$$

$$h'(m) = -\frac{1}{2-m}, \quad h'(1) = -1$$

$$h''(m) = \frac{1}{(2-m)^2}, \quad h''(1) = 1$$

$$h(m) = -(m-1) - \frac{(m-1)^2}{2} + (m-1)^2 \mathcal{E}(m)$$

$$f(m) = g(m) \cdot h(m) = -(m-1) - \frac{(m-1)^2}{2} + (m-1)^2 \mathcal{E}(m)$$

$$g(m) = \frac{m-1}{\ln m} \text{ at } m_0 = 1, \text{ order 2}$$

method change of variable

let $h = m-1$ then $m = h+1$

$$g(m) = \frac{m-1}{\ln m} \text{ then } g(h) = \frac{h}{\ln(h+1)}$$

we know that the LD of

$$\ln(1+h) \text{ at } h_0 = 0 \text{ to order 2 is}$$

$$\ln(1+h) = -h + \frac{h^2}{2} + h^2 \mathcal{E}(h)$$

then the LD of $\left(\frac{h}{\ln(1+h)} \right)$

$$\text{is } -$$

$$\ln(1+h) = h - \frac{h^2}{2} + h^2 \mathcal{E}(h)$$

$$\text{so, } g(h) = \frac{h}{h - \frac{h^2}{2} + h^2 \mathcal{E}(h)}$$

$$= \frac{1}{1 - \frac{h}{2} + h \mathcal{E}(h)}$$

$$\text{Let } u(h) = \frac{h}{2}, \text{ then } h \rightarrow 0$$

$$g(h) = \frac{1}{1 - u(h)} \quad u(h) \rightarrow 0$$

we know that the LD of $\frac{1}{1+u}$

$$\text{at } u_0 = 0 \text{ to order 2 is}$$

$$\frac{1}{1+u} = 1 - u + u^2 + u^2 \mathcal{E}(u)$$

$$\text{then } \frac{1}{1-u} = 1 + u + u^2 + u^2 \mathcal{E}(u)$$

$$\text{so } g(h) = 1 + u(h) + u^2(h) + u^2(h) \mathcal{E}(h)$$

$$u(h) = \frac{h}{2} = \frac{(m-1)}{2}$$

$$g(m) = \frac{1}{2} + \frac{(m-1)}{2} + \frac{(m-1)^2}{4}$$

$$\rho(m) = \frac{3\sqrt{m-1}}{m-m^2}$$

we are going to calculate it's LD at $m_0 = 1$ to order 3

$m_0 \notin D_e$, by putting $t = 1-m$, so $m = 1-m$.

$$\rho(t) = \frac{3\sqrt{t}}{(t+1)-t^2+t^2} = \frac{3\sqrt{t}}{1-t} = \frac{1}{1-t} t^{-2/3}, \quad \rho(0) =$$

$$\rho(t) = t^{-2/3} \left(\frac{1}{1-t} \right), \quad \rho(t) = (-2/3) (1+t+t^2+t^3+t^4+t^5+\dots) + \rho(t^3)$$

$$\rho(t) = t^{-2/3} + t^{1/3} + t^{4/3} + t^{7/3} + \dots \quad \text{Then}$$

$$\rho(m) = -(1-m)^{-2/3} = -(1-m)^{2/3} + \theta(1-m)^{5/3}$$

Exos:

$$f(x) = x e^{\frac{2x}{1+x}} = x e^{\frac{2}{1+\frac{1}{x}}}$$

Let $y = \frac{1}{x}$. when $x \rightarrow \pm \infty$
s.t. $y \rightarrow 0$

$$f\left(\frac{1}{y}\right) = \frac{e^{\frac{2}{y}} \cdot \frac{1}{1+y}}{y} = \frac{1}{y} e^{\frac{2}{1+y}}$$

Notice that,

$$\frac{1}{1+y} = 1 - y + y^2 + o(y^2)$$

$$\text{and } e^{\frac{2}{1+y}} = e^{1-y+y^2+o(y^2)} = e^1 \cdot e^{-y+y^2+o(y^2)} = e \cdot e^{-y+y^2+o(y^2)}$$

$$e^u = 1 + u + \frac{u^2}{2!} + o(u^2)$$

Let $u(y) = \frac{2}{1+y}$ $u(0) = 0$

$$e^{-y+y^2+o(y^2)} = 1 - y + y^2 + \frac{1}{2}y^2 + o(y^2) = 1 - y + \frac{3}{2}y^2 + o(y^2)$$

$$\text{So } \left(e^{-y+y^2+o(y^2)} \right)^2 = 1 - 2y + 4y^2 + o(y^2)$$

Thus,

$$f(x) = x \cdot \left(e^2 \left(1 - \frac{2}{x} + 4\frac{1}{x^2} + o\left(\frac{1}{x^2}\right) \right) \right)$$

Therefore

$$f(x) = e^2 x - 2e^2 + 4e^2 \frac{1}{x} + o\left(\frac{1}{x}\right)$$

we deduce that,

The equation of the asymptote is $y = e^2(x - 2)$

$$0 - y + \frac{3}{2} =$$

$$\left(1 - y + \frac{3}{2} \right)$$

$$x - y + \frac{3}{2} = y + y^2$$

Ex 6:

$$\lim_{x \rightarrow 0} \frac{2x(1-\cos x)}{(e^x - 1) \log x} \lim_{x \rightarrow 0} \exp\left(\frac{1}{x} \ln(1+x)\right)$$

$$1 - \cos x = 1 - \left(1 - \frac{x^2}{2} + o(x^2)\right) = \frac{x^2}{2} + o(x^2)$$

$$e^x \left(\frac{x^2}{2} + o(x^2)\right) = \frac{x^2}{2} + o(x^2)$$

$$(e^x - 1) \log x = \left(x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)\right) \left(x - \frac{x^3}{3} + o(x^3)\right)$$

$$= x^2 + \frac{x^3}{2} + o(x^3)$$

$$\lim_{x \rightarrow 0} \frac{x^2 + x^3 \varepsilon(x)}{x^2 + x^3 \varepsilon(x)} = \lim_{x \rightarrow 0} \frac{x + x \varepsilon(x)}{\frac{x}{2} + 1 + x \varepsilon(x)} = \frac{0}{1} = 0$$

$$(a+b)^3 = a^3 + b^3 + 3a^2b + 3ab^2$$

$$\lim_{x \rightarrow 0} \frac{e^x - \exp(2-x)}{x - \sin x} = \frac{0}{0}$$

we have:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)$$

$$\sin x = x - \frac{x^3}{6} + o(x^3)$$

$$\exp(u) = 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + o(x^3)$$

$$\text{Put: } u = \sin x = x - \frac{x^3}{6} + o(x^3)$$

$$\exp(2-x) = 1 + x - \frac{x^2}{2} + \frac{1}{6} \left(x - \frac{x^3}{6}\right)^2 + \frac{1}{6} \left(x - \frac{x^3}{6}\right)^3 + o(x^3)$$

$$= 1 + x - \frac{x^2}{2} + \frac{1}{2}x^2 + \frac{x^3}{3} + o(x^3)$$

$$= 1 + x + \frac{x^2}{2} + o(x^2)$$

$$\lim_{x \rightarrow 0} \frac{e^x - \exp(2-x)}{x - \sin x} = \lim_{x \rightarrow 0} \frac{1 - \frac{x^3}{6} + o(x^3)}{\frac{x^3}{6} + o(x^3)} = 1$$

$$= C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + \dots$$

$$e^{-y} L(y)$$

$$\Rightarrow \lim_{x \rightarrow +\infty} \frac{2}{\pi} \left(\frac{\pi}{2} - \arctg\left(\frac{1}{x}\right) \right) - 1$$

$$= \lim_{x \rightarrow +\infty} -\frac{2}{\pi} \left(\arctg\left(\frac{1}{x}\right) \right)$$

$$\text{let } t = \frac{1}{x} \quad t \rightarrow 0$$

$$\Rightarrow \lim_{t \rightarrow 0} -\frac{2}{\pi} \left(\arctg t \right)$$

$$= \lim_{t \rightarrow 0} -\frac{2}{\pi} \left(t - \frac{t^3}{3} + o(t^5) \right) = 0$$

$$(f) \lim_{u \rightarrow 1} \frac{1}{(u-1)^{n-1}} (\ln u)^{n-1} = \lim_{u \rightarrow 1} \exp \left((n-1) \ln(\ln u) \right)$$

$$= \lim_{t \rightarrow 0} \exp \left(t \ln(\ln(t+1)) \right)$$

$$= \lim_{t \rightarrow 0} \exp \left(t \ln \left(t - \frac{t^2}{2} + o(t^2) \right) \right)$$

Ex 6:

$$\lim_{x \rightarrow 0} x \arctan\left(\frac{1}{1-x}\right) \quad x = \frac{1}{x} \quad \begin{matrix} x \rightarrow \infty \\ x \rightarrow 0 \end{matrix}$$

$$\lim_{x \rightarrow 0} x \arctan\left(\frac{1}{1-x}\right) = \lim_{x \rightarrow 0} \frac{1}{x} \arctan\left(\frac{1}{1-\frac{1}{x}}\right) = \lim_{x \rightarrow 0} \frac{1}{x} \arctan\left(\frac{x}{x-1}\right)$$

$$\frac{1}{x-1} = \frac{-1}{1-x} = -(1+x+o(x))$$

$$\lim_{x \rightarrow 0} x \arctan\left(\frac{1}{1-x}\right) = \lim_{x \rightarrow 0} \frac{1}{x} \arctan\left(-x - x^2 + o(x^2)\right)$$

$$= \lim_{x \rightarrow 0} \frac{1}{x} (-x - x^2 + o(x^2)) = \lim_{x \rightarrow 0} -1 - x + o(x) = -1$$

Exos:

Determine the equation of the asymptote of curve.

$$g(x) = \sqrt[3]{x^3 + x + 1} = x \cdot \sqrt[3]{1 + \underbrace{\left(\frac{1}{x^2} + \frac{1}{x^3}\right)}_u}$$

$$\text{Let } y = \frac{1}{x}. \quad \text{If } x \rightarrow \pm\infty \\ y \rightarrow 0.$$

$$y = ax + b$$

We have, the LD(0) of $\sqrt[3]{1+y}$ is given by.

$$\sqrt[3]{1+y} = 1 + \frac{1}{3}y + o(y).$$

So

$$\sqrt[3]{1 + \left(\frac{1}{x^2} + \frac{1}{x^3}\right)} = 1 + \frac{1}{3}\left(\frac{1}{x^2} + \frac{1}{x^3}\right) + o\left(\frac{1}{x^2} + \frac{1}{x^3}\right).$$

Then.

$$g(x) = \underbrace{x}_{\text{asymptote}} + \frac{1}{3} \frac{1}{x} + \frac{1}{3} \frac{1}{x^2} + o\left(\frac{1}{x^2}\right).$$